NUDGE

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ABSTRACT

1 Method

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $(\mathcal{F}_t)_{t\geq 0}$. We consider the reference (true) dynamics governed by the following McKean-Vlasov stochastic differential equation

$$\mathrm{d}\mathbf{X}_t = \mathbf{b}_{\mathrm{true}}(\mathbf{X}_t, \mu_t)\mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t \quad \mathbf{X}_0 \sim \mu_0$$

where $\mu_t = \mathrm{Law}(\mathbf{X}_t) \in \mathcal{P}_2(\mathbb{R}^d)$ denotes the probability distribution of the true process, $\mathbf{X}_t \in \mathbb{R}^d$ is the state of the reference particle, $\mathbf{W}_t$ is a standard d-dimensional Brownian motion, $\mathbf{b}_{\mathrm{true}} : \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}^d$ is the true drift term and $\Sigma \in \mathbb{R}^{d\times d}$ is the diffusion coefficient (assumed constant for simplicity). In practice, the true drift $\mathbf{b}_{\mathrm{true}}$ is often unknown; we assume access to an approximate model drift $\mathbf{b}_{\mathrm{model}}$, which deviates from the truth as:

$$\mathbf{R}(\mathbf{x}, \mu) := \mathbf{b}_{\mathrm{model}}(\mathbf{x}, \mu) - \mathbf{b}_{\mathrm{true}}(\mathbf{x}, \mu).$$

The approximate system $\mathbf{Y}_t$ would follow:

$$\mathrm{d}\mathbf{Y}_t = \mathbf{b}_{\mathrm{model}}(\mathbf{Y}_t, \nu_t)\mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t,$$

where $\nu_t = \mathrm{Law}(\mathbf{Y}_t)$. Due to errors on initialization and inherent model bias, the discrepancy between the true and approximate systems tends to accumulate over time. However, in many practical applications, we have access to partial information via an observation operator acting on the system's law. For example, in this paper, we assume that observations are given by

$$\mathcal{O}_h(\mu_t)(\mathbf{z}) = \int K_h(\mathbf{z} - \mathbf{x})\mu_t(\mathrm{d}\mathbf{x}),$$

which can be approximated by partial observation of the location $\{\mathbf{X}_t^{\mathrm{obs},k,M}\}_{k=1}^M$. To reduce the discrepancy induced by initialization and model error, we augment the approximate dynamics with a feedback control ("nudging") term that drives ν_t toward consistency with the observations. Concretely, we define the assimilated particle dynamics

$$\mathrm{d}\mathbf{Y}_t = \mathbf{b}_{\mathrm{model}}(\mathbf{Y}_t, \nu_t)\,\mathrm{d}t + \mathbf{u}_{\mathrm{nud}}(\mathbf{Y}_t, \nu_t, \mu_t^{\mathrm{obs}})\,\mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t$$

To reduce the discrepancy induced by initialization and model error, data assimilation method [1] was proposed to augment the approximate dynamics with a nudging term that drives ν_t toward consistency with the observations. Concretely, we define the assimilated particle dynamics

$$\mathrm{d}\mathbf{Y}_t = \mathbf{b}_{\mathrm{model}}(\mathbf{Y}_t, \nu_t)\,\mathrm{d}t + \mathbf{u}_{\mathrm{nud}}(\mathbf{Y}_t, \nu_t, \mu_t^{\mathrm{obs}})\,\mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t$$

A natural choice is to select $\mathbf{u}_{\mathrm{nud}}$ as the Wasserstein gradient flow of a quadratic observation misfit functional. Define the cost functional as

$$J(\nu, \mu^{\mathrm{obs}}) = \frac{1}{2}\int_{\mathbb{R}^d} \left|(K_h * \nu)(\mathbf{x}) - (K_h * \mu^{\mathrm{obs}})(\mathbf{x})\right|^2 \mathrm{d}\mathbf{x},$$

Computing the Fèchet derivative of J with respect to the measure ν, denoted as $\frac{\delta J}{\delta \nu}$, yields:

$$\frac{\delta J}{\delta \nu}(\mathbf{x}) = \left(\tilde{K}_h * (\nu - \mu^{\mathrm{obs}})\right)(\mathbf{x}),$$

where $\tilde{K}_h(\mathbf{z}) = K_h * K_h$ under the assumption that $\tilde{K}$ is symmstric. In the Wasserstein geometry, the gradient flow corresponds to a vector field defined by the negative gradient of this first variation. Thus, introducing a nudging intensity $\lambda > 0$, we set:

$$\mathbf{u}_{\mathrm{nud}}(\mathbf{y}, \nu, \mu^{\mathrm{obs}}) = -\lambda \nabla_{\mathbf{y}} \left(\frac{\delta J}{\delta \nu}(\mathbf{y}) \right) = -\lambda \nabla \left(\tilde{K}_h * (\nu - \mu^{\mathrm{obs}}) \right)(\mathbf{y}).$$

Therefore, we can approximate the assimilated approximation process by finite particles as

$$\mathrm{d}\mathbf{Y}_t^{i,N} = \mathbf{b}_{\mathrm{model}}(\mathbf{Y}_t^{i,N}, \nu_t^N)\mathrm{d}t - \lambda \left[\nabla \left(\tilde{K}_h * \nu_t^N \right)(\mathbf{Y}_t^{i,N}) - \nabla \left(\tilde{K}_h * \mu_t^{\mathrm{obs}} \right)(\mathbf{Y}_t^{i,N}) \right] \mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t^i,$$

where $\nu_t^N = \frac{1}{N}\sum_{j=1}^N \delta_{\mathbf{Y}_t^{j,N}}$ is the empirical measure of the system. Expanding the convolution terms, the explicit interaction dynamics are given by:

$$\mathrm{d}\mathbf{Y}_t^{i,N} = \mathbf{b}_{\mathrm{model}}\left(\mathbf{Y}_t^{i,N}, \nu_t^N\right)\mathrm{d}t - \lambda \left(\frac{1}{N}\sum_{j=1}^N \nabla \tilde{K}_h(\mathbf{Y}_t^{i,N} - \mathbf{Y}_t^{j,N}) - \frac{1}{M}\sum_{k=1}^M \nabla \tilde{K}_h(\mathbf{Y}_t^{i,N} - \mathbf{X}_t^{\mathrm{obs},k,M}) \right) \mathrm{d}t + \Sigma \mathrm{d}\mathbf{W}_t^i$$

Formally, the Law of $\mathbf{Y}_t$, denoted by ν_t^{ass}, satisfies the following nonlinear Fokker-Planck equation:

$$\partial \nu_t = -\nabla \cdot (\nu_t \mathbf{b}_{\mathrm{model}}(\mathbf{x}, \nu_t)) + \nabla \cdot (A \nabla \nu_t) + \lambda \nabla \cdot (\nu_t \nabla (K_h * r)),$$

where $A = \Sigma\Sigma^\top$ and $r = K_h * (\nu_t - \mu_t^{\mathrm{obs}})$. Notice that the true mean field dynamics is given by

$$\partial \mu_t = -\nabla \cdot (\mu_t \mathbf{b}_{\mathrm{true}}(\mathbf{x}, \mu_t)) + \nabla \cdot (A \nabla \mu).$$

Subtracting the true dynamics from the assimilated dynamics, the error $v_t = \nu_t - \mu_t$ satisfies

$$\partial_t v_t = -\nabla \cdot (\nu_t \mathbf{b}_{\mathrm{model}}(\mathbf{x}, \nu_t) - \mu_t \mathbf{b}_{\mathrm{true}}(\mathbf{x}, \mu_t)) + \nabla \cdot (A \nabla v_t) + \lambda \nabla \cdot (\nu_t \nabla (K_h * r))$$

Assumption 1 (Regularity, stability, and bounded error sources)**.** Let Ω be a periodic domain (e.g. $\mathbb{T}^d$). Assume the following hold.

(A1) Bounded model drift. There exists a constant $B_{\mathrm{model}} > 0$ such that for all $\rho \in \mathcal{P}_2(\Omega)$,

$$\|\mathbf{b}_{\mathrm{model}}(\cdot, \rho)\|_{L^\infty(\Omega)} \le B_{\mathrm{model}}, \qquad \|\nabla \cdot \mathbf{b}_{\mathrm{model}}(\cdot, \rho)\|_{L^\infty(\Omega)} \le B_{\mathrm{model}}.$$

(A2) Measure-Lipschitz true drift. There exists a constant $L_\mu > 0$ such that for all $x \in \Omega$ and all $\rho, \rho' \in \mathcal{P}_2(\Omega)$,

$$\|\mathbf{b}_{\mathrm{true}}(x, \rho) - \mathbf{b}_{\mathrm{true}}(x, \rho')\| \le L_\mu \|\rho - \rho'\|.$$

(A3) Bounded densities and positivity. The true density is uniformly bounded, and the assimilated density is uniformly bounded below: there exist constants $\bar{\rho} > 0$ and $\underline{\nu} > 0$ such that, for all $t \ge 0$,

$$\|\mu_t\|_{L^\infty(\Omega)} \le \bar{\rho}, \qquad \nu_t(x) \ge \underline{\nu} \text{ for a.e. } x \in \Omega.$$

(A4) Uniformly bounded model bias. There exists a constant $\Delta > 0$ such that

$$\sup_{t \ge 0} \|\mathbf{b}_{\mathrm{model}}(x, \rho) - \mathbf{b}_{\mathrm{true}}(x, \rho)\|^2 \le \Delta^2.$$

(A5) Symmetric Kernel. The kernel K_h is symmetric and $W^{1,1}$, so convolution is self-adjoint on $L^2(\Omega)$ and commutes with derivatives.

(A6)Smoothing Error Estimate. The smoothing kernel $\tilde{K}_h = K_h * K_h$ approximates the identity on the gradient space. Specifically, there exists an error function $\delta(h)$ with $\lim_{h \to 0} \delta(h) = 0$ such that for any $v \in H^1(\Omega)$:

$$\|\nabla v - \nabla(\tilde{K}_h * v)\|_{L^2(\Omega)} \le \delta(h) \|\nabla v\|_{L^2(\Omega)}.$$

Theorem 1.1. *Assume Assumption 1 holds and that $\int_\Omega v_0\,dx = 0$. Let C_P denote the Poincaré constant on Ω (for zero-mean functions). We also assume $\mu_t^{\mathrm{obs}} = \mu_t$. Let $C_{\mathrm{adv}} = B_{model} + \bar{\rho} L_\mu$ denote the coefficient of the advection instability. Suppose the observation resolution h and the nudging intensity λ are chosen to satisfy the following conditions: The observation scale h is sufficiently small such that the smoothing error is dominated by the density lower bound:*

$$\delta(h) \le \frac{\underline{\nu}}{2\bar{\rho}}.$$

The nudging intensity λ is sufficiently large to overcome the advection instability:

$$\lambda > \frac{2}{\underline{\nu}}\left(\frac{C_{\mathrm{adv}}^2 C_P}{\kappa} - \kappa\right).$$

Then, the squared L^2-error $V(t) = \frac{1}{2}\|\nu_t - \mu_t\|_{L^2}^2$ decays exponentially to a neighborhood of the origin. Specifically, there exists a rate $\alpha > 0$:

$$\frac{d}{dt}V(t) \le -\alpha V(t) + \frac{\bar{\rho}^2}{\kappa}\Delta^2.$$

Consequently, for all $t \ge 0$:

$$V(t) \le e^{-\alpha t}V(0) + \frac{\bar{\rho}^2}{\alpha\kappa}\Delta^2,$$

implying that the long-time error is bounded by a bias floor proportional to Δ^2.

Proof. For the L^2 error energy $V(t) = \frac{1}{2}\|v_t\|^2$, the time derivative is given by:

$$V'(t) = \langle v_t, \partial_t v_t\rangle.$$

Using the coercivity of the diffusion matrix A, we have:

$$\langle v_t, \nabla\cdot(A\nabla v_t)\rangle \le -\kappa\|\nabla v_t\|^2.$$

Let $C_{\mathrm{adv}} = B_{\mathrm{model}} + \bar{\rho}L_\mu$. Using Young's Inequality with a parameter $\epsilon > 0$ (to be chosen later), we estimate the nonlinear terms:

$$\begin{aligned}
&\langle \nabla v_t, \nu_t \mathbf{b}_{\mathrm{model}}(\mathbf{x},\nu_t) - \mu_t \mathbf{b}_{\mathrm{true}}(\mathbf{x},\mu_t)\rangle \\
&\le B_{\mathrm{model}}\|\nabla v_t\|\|v_t\| + \bar{\rho}L_\mu\|\nabla v_t\|\|v_t\| + \bar{\rho}\|\nabla v_t\|\Delta \\
&= C_{\mathrm{adv}}\|\nabla v_t\|\|v_t\| + \bar{\rho}\|\nabla v_t\|\Delta \\
&\le \frac{\epsilon}{2}\|\nabla v_t\|^2 + \frac{C_{\mathrm{adv}}^2}{2\epsilon}\|v_t\|^2 + \frac{\epsilon}{2}\|\nabla v_t\|^2 + \frac{\bar{\rho}^2}{2\epsilon}\Delta^2 \\
&= \epsilon\|\nabla v_t\|^2 + \frac{C_{\mathrm{adv}}^2}{2\epsilon}\|v_t\|^2 + \frac{\bar{\rho}^2}{2\epsilon}\Delta^2.
\end{aligned}$$

We treat the nudging term as a perturbation of the identity. Using the property $\nabla\psi_t = \nabla(\tilde{K}_h * v_t)$ and the assumption that the smoothing error satisfies $\|\nabla v_t - \nabla(\tilde{K}_h * v_t)\| \le \delta(h)\|\nabla v_t\|$, we have:

$$\begin{aligned}
\langle v_t, \lambda\nabla\cdot(\nu_t\nabla\psi_t)\rangle &= -\lambda\langle \nu_t\nabla v_t, \nabla\psi_t\rangle \\
&= -\lambda\langle \nu_t\nabla v_t, \nabla v_t\rangle - \lambda\langle \nu_t\nabla v_t, \nabla\psi_t - \nabla v_t\rangle \\
&\le -\lambda\underline{\nu}\|\nabla v_t\|^2 + \lambda\bar{\rho}\|\nabla v_t\|\|\nabla\psi_t - \nabla v_t\| \\
&\le -\lambda\underline{\nu}\|\nabla v_t\|^2 + \lambda\bar{\rho}\delta(h)\|\nabla v_t\|^2 \\
&= -\lambda(\underline{\nu} - \bar{\rho}\delta(h))\|\nabla v_t\|^2.
\end{aligned}$$

Combining the estimates above, we obtain:

$$V'(t) \le \left(-\kappa - \lambda(\underline{\nu} - \bar{\rho}\delta(h)) + \epsilon\right)\|\nabla v_t\|^2 + \frac{C_{\mathrm{adv}}^2}{2\epsilon}\|v_t\|^2 + \frac{\bar{\rho}^2}{2\epsilon}\Delta^2.$$

To ensure stability, we first choose the observation resolution h sufficiently small such that $\delta(h) \le \frac{\underline{\nu}}{2\bar{\rho}}$, which implies $\underline{\nu} - \bar{\rho}\delta(h) \ge \frac{\underline{\nu}}{2}$. We then set $\epsilon = \frac{\kappa}{2}$ to absorb the advection gradient term. The estimate becomes:

$$V'(t) \le -\left(\frac{\kappa}{2} + \frac{\lambda\underline{\nu}}{2}\right)\|\nabla v_t\|^2 + \frac{C_{\mathrm{adv}}^2}{\kappa}\|v_t\|^2 + \frac{\bar{\rho}^2}{\kappa}\Delta^2.$$

Finally, applying the Poincaré inequality $\|\nabla v_t\|^2 \ge C_P^{-1}\|v_t\|^2$, we get:

$$V'(t) \le -\left[\frac{1}{C_P}\left(\frac{\kappa}{2} + \frac{\lambda\underline{\nu}}{2}\right) - \frac{C_{\mathrm{adv}}^2}{\kappa}\right]\|v_t\|^2 + \frac{\bar{\rho}^2}{\kappa}\Delta^2.$$

$\square$

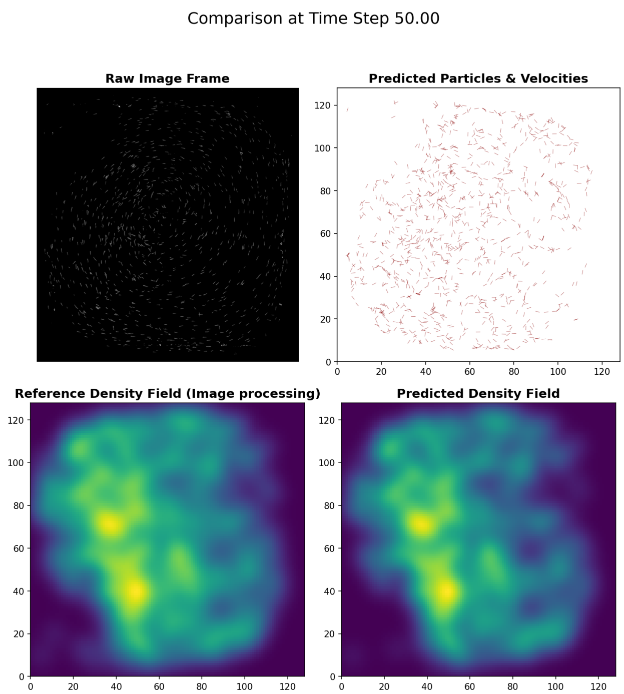

Figure 1: The comparison of the predicted fish position and reference data

2 Numerical Result

We applied the proposed mean-field data assimilation framework to trajectory data obtained from experiments with approximately $N \approx 1000$ animals (fish) swimming in a quasi-two-dimensional tank [2]. The dataset consists of discrete-time position measurements $\{\mathbf{X}_i(t_n)\}$. These observations are spatially and temporally incomplete due to visual occlusions and tracking dropouts inherent to the image processing pipeline. The domain Ω was discretized using a uniform grid with spatial resolution Δx, and the density was approximated using a kernel density estimator to ensure regularity consistent with Assumption 1:

$$\mu_{obs}(x, t_n) = \frac{1}{N} \sum_{i=1}^{N} K_\sigma(x - X_i(t_n)), \tag{1}$$

where K_h is a Gaussian smoothing kernel with bandwidth h. We train a model using the previous method [**?**] and data assimilation by the current one, the numerical result it shown in Figure 1

References

[1] H. Bessaih, E. Olson, and E. S. Titi. Continuous data assimilation with stochastically noisy data. *Nonlinearity*, 28(3):729, 2015.

[2] T. Walter and I. D. Couzin. Trex, a fast multi-animal tracking system with markerless identification, and 2d estimation of posture and visual fields. *Elife*, 10:e64000, 2021.

We consider the Vlasov–Fokker–Planck equation

$$\partial_t f + \mathbf{v}\cdot\nabla_{\mathbf{x}} f + \big(\mathbf{E}(\mathbf{x}) + \mathbf{F}(t,\mathbf{x})\big)\cdot\nabla_{\mathbf{v}} f = \frac{\sigma^2}{2}\Delta_{\mathbf{v}} f, \tag{2}$$

where the self-consistent field $\mathbf{E}$ is determined by

$$\nabla_{\mathbf{x}}\cdot\mathbf{E}(t,\mathbf{x}) = 4\pi\,\rho(t,\mathbf{x}), \qquad \rho(t,\mathbf{x}) := \int_{\mathbb{R}^d} f(t,\mathbf{x},\mathbf{v})\,d\mathbf{v}, \tag{3}$$

and $\mathbf{F}(t,\mathbf{x})$ denotes a prescribed external forcing. Equation (2) is the forward Kolmogorov equation associated with the SDE

$$d\mathbf{X}_t = \mathbf{V}_t\,dt, \qquad d\mathbf{V}_t = \big(\mathbf{E}(\mathbf{X}_t) + \mathbf{F}(t,\mathbf{X}_t)\big)\,dt + \sigma\,d\mathbf{W}_t. \tag{4}$$

Throughout, we assume the model is exact and that the only source of discrepancy arises from imperfect initialization. The available observation is the macroscopic density $\rho(t,\mathbf{x})$ defined in (3).

To incorporate the density observation, we introduce a nudged copy of (4):

$$d\mathbf{Y}_t = \mathbf{U}_t\,dt - \mathbf{u}_{\mathrm{nud}}\big(\mathbf{Y}_t,\tilde\rho_t,\rho_t\big)\,dt, \qquad d\mathbf{U}_t = \big(\mathbf{E}(\mathbf{Y}_t) + \mathbf{F}(t,\mathbf{Y}_t)\big)\,dt + \sigma\,d\mathbf{W}_t, \tag{5}$$

where $\tilde\rho_t$ denotes the spatial density induced by $\mathbf{Y}_t$, i.e.,

$$\tilde\rho(t,\mathbf{x}) := \mathrm{Law}(\mathbf{Y}_t)(d\mathbf{x}), \qquad \text{equivalently} \qquad \tilde\rho(t,\mathbf{x}) = \int_{\mathbb{R}^d}\tilde f(t,\mathbf{x},\mathbf{v})\,d\mathbf{v},$$

with $\tilde f$ the phase-space density of $(\mathbf{Y}_t,\mathbf{U}_t)$. The nudging term $\mathbf{u}_{\mathrm{nud}}(t,\mathbf{x}) = \lambda\,\nabla_{\mathbf{x}}\big(K_h * r(t,\cdot)\big)(\mathbf{x})$ and $r := K_h*(\tilde\rho-\rho)$ is designed so that $\tilde\rho$ is driven toward the observed density ρ. Define that $\omega = \tilde f - f$, then we have

$$\partial_t\omega + \mathbf{v}\cdot\nabla_{\mathbf{x}}\omega + \tilde{\mathbf{E}}(t,\mathbf{x})\cdot\nabla_{\mathbf{v}}\tilde f - \mathbf{E}(t,\mathbf{x})\cdot\nabla_{\mathbf{v}} f + \mathbf{F}(t,\mathbf{x})\cdot\nabla_v\omega = \frac{\sigma^2}{2}\Delta_{\mathbf{v}}\omega + \lambda\nabla_{\mathbf{x}}\cdot(\tilde f\nabla_{\mathbf{x}}(K_h * r))$$